A REPORT

ON

INTELLIPRINT NEXUS

AI-Powered Intelligent Printing, Queue Optimization & Printer Management Platform

Submitted by

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Under the guidance of,

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

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PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on **" INTELLIPRINT NEXUS "** being submitted by " **SUFIYAN SAMEER SHAIKH** " bearing roll number " **20241CSE1091** " in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering (Computer Science and Engineering)** is a bonafide work carried out under our supervision.

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I hereby declare that the work, which is being presented in the report entitled " **INTELLIPRINT NEXUS "** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering (Computer Science and Engineering),** is a record of my own investigations carried under the guidance of **Ms. Vineetha B,** Assistant Professor, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

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First of all, I indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this internship on time.

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ABSTRACT

Campus printing services in educational institutions often depend on manual document submission, printer selection, payment confirmation, queue handling, and physical document collection. During examinations, assignment submissions, project deadlines, and short class breaks, high printing demand can result in long queues, crowding, inefficient printer utilization, and difficulty in tracking print jobs. Traditional workflows also provide limited information about printer availability, expected waiting time, workload, and document quality before printing. **IntelliPrint Nexus** is proposed as an AI-powered intelligent campus printing and queue optimization platform that integrates Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, intelligent queue management, printer management, payment processing, and automated print-job tracking into a unified digital workflow. The platform enables authenticated users to upload PDF documents, configure printing preferences, analyze document quality, complete payment, monitor print-job progress, and collect completed documents using a unique Print ID and QR Code.

A key component of IntelliPrint Nexus is the AI-based Document Quality Analysis module, which evaluates uploaded documents using Computer Vision and OpenCV-based image processing. Parameters such as sharpness, brightness, contrast, blur, orientation, skewness, and readability are considered to identify potential quality issues. A Convolutional Neural Network approach using MobileNetV2 and transfer learning is incorporated for document quality assessment. The system also implements intelligent queue and printer management by considering page count, queue length, printer availability, printer speed, workload, and job priority. Machine Learning approaches, including Random Forest and XGBoost, support prediction-related tasks such as waiting-time estimation and printer recommendation.

The platform integrates payment processing, digital receipt generation, real-time print-job tracking, and QR-based document collection. The administrative component supports monitoring of students, print jobs, printers, payments, queues, notifications, reports, and system analytics. Its modular architecture separates application views, authentication, database management, AI processing, printer management, queue management, and supporting utilities.

The platform is implemented using Python and Streamlit for the application interface, SQLite and SQLAlchemy for database management, TensorFlow and Keras for deep-learning integration, OpenCV for computer vision and image processing, and Scikit-learn and XGBoost for machine-learning tasks. Additional libraries support PDF processing, QR Code generation, receipt generation, authentication, notifications, and printer communication. Overall, IntelliPrint Nexus demonstrates the application of AI and modern software engineering to a real-world university printing problem by integrating document analysis, print configuration, payment, queue optimization, printer recommendation, tracking, and QR-based collection while aiming to reduce waiting time and congestion, improve printer utilization, and enhance the campus printing experience.

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CHAPTER -1

INTRODUCTION

1.1 About the Internship Program

The Summer Internship Program on Artificial Intelligence and High-Performance Computing was conducted to provide students with practical exposure to advanced computing technologies and real-world software development practices. The program focused on emerging areas such as Artificial Intelligence, Machine Learning, Generative AI, Large Language Models (LLMs), cloud computing, and High-Performance Computing (HPC). Through a combination of technical sessions, hands-on activities, and project-based learning, participants were encouraged to develop innovative solutions that address real-world challenges using modern technologies.

The increasing adoption of Artificial Intelligence across industries has transformed the way organizations develop software, analyze data, automate processes, and deliver intelligent services. As a result, there is a growing demand for professionals with expertise in AI-driven systems, cloud platforms, scalable architectures, and modern development frameworks. The internship program served as a platform to bridge the gap between academic learning and industry requirements by enabling students to apply theoretical concepts in practical development environments.

The program also encouraged independent problem-solving, teamwork, innovation, and technical research. Students were given the opportunity to explore current trends in AI and software engineering while developing projects that demonstrate practical applications of emerging technologies. This exposure helped in understanding the complete software development lifecycle, from problem identification and system design to implementation, testing, and deployment.

1.2 About the Project – IntelliPrint Nexus

IntelliPrint Nexus is an AI-powered smart campus printing and queue-management platform designed to modernize the conventional printing process in educational institutions. The system provides a centralized digital platform through which students can upload documents, configure printing requirements, analyze document quality, select suitable printers, complete the payment process, track print jobs, and collect completed documents using QR Code-based identification. The project integrates Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and modern software engineering techniques to provide an efficient and organized printing workflow.

The platform addresses several limitations associated with conventional campus printing, where students may need to manually submit documents, communicate printing requirements, wait in queues, verify payments, and repeatedly check the status of their print requests. IntelliPrint Nexus aims to reduce these manual activities by providing a unified digital workflow from document submission to final collection.

A major feature of the project is the AI-assisted document quality analysis module. The system uses Computer Vision and image-processing techniques to examine uploaded documents before printing.

OpenCV-based processing and a MobileNetV2-based Deep Learning approach with transfer learning are incorporated to assess characteristics such as sharpness, brightness, contrast, blur, orientation, alignment, and readability. This analysis helps identify potential document-quality issues before printing and can reduce unnecessary reprints and paper wastage.

The project also incorporates intelligent printer recommendation and queue management. Information such as printer availability, queue length, printer speed, workload, document page count, and job priority can be considered for estimating waiting time and recommending a suitable printer. This enables users to make more informed printing decisions while supporting better distribution of printing workloads.

The complete workflow includes user authentication, document upload, print configuration, AI-based analysis, printer recommendation, payment processing, queue assignment, print-status tracking, notifications, and QR Code-based document collection. The system also provides separate student and administrative interfaces. Students can manage their printing activities, view print history, monitor queues, access receipts, and collect completed documents, while administrators can monitor users, printers, print jobs, payments, queues, notifications, and system information.

Overall, IntelliPrint Nexus demonstrates the practical application of AI and intelligent automation to campus printing. By integrating document quality assessment, printer recommendation, queue management, digital payment workflow, real-time tracking, and QR-based collection into a single platform, the project aims to reduce waiting time and manual intervention, improve printer utilization, minimize resource wastage, and provide students with a more convenient and transparent printing experience.

1.3 Problem Statement

Conventional campus printing services often rely on manual processes for document submission, print configuration, payment verification, printer selection, queue management, and document collection. During periods of high demand, such as examinations, assignment deadlines, and project submissions, these processes can result in long queues, increased waiting times, higher operator workload, and inefficient utilization of available printers. Students may also have limited visibility into the status of their print requests.

Additionally, conventional printing systems generally lack automated document-quality assessment, intelligent printer selection, waiting-time estimation, and integrated queue tracking. This can lead to poor-quality prints, unnecessary reprinting, resource wastage, and delays. Therefore, there is a need for an integrated and intelligent printing platform that can streamline the printing workflow, improve printer utilization, reduce waiting time, and provide students with better visibility and control over their print requests.

CHAPTER - 2

OBJECTIVES

The primary objective of IntelliPrint Nexus is to develop an AI-powered intelligent campus printing platform that improves the efficiency, transparency, and user experience of university printing services. The system aims to reduce manual intervention in printing operations by integrating document quality analysis, intelligent queue management, printer recommendation, online print management, payment processing, and real-time print-job tracking within a unified digital platform. The proposed system focuses on applying Artificial Intelligence, Machine Learning, Deep Learning, and Computer Vision to selected stages of the printing workflow. It is designed to assist in document quality assessment, estimate printing waiting time, utilize available printer information, and support efficient management of print requests through an organized digital workflow.

The platform also aims to provide students with relevant information throughout the printing process, including document quality results, printer availability, estimated waiting time, queue position, payment status, and print-job progress. From the administrative perspective, the system provides centralized visibility of users, print requests, printers, queues, payments, notifications, and operational information. By integrating these capabilities, IntelliPrint Nexus seeks to reduce unnecessary manual activities, improve utilization of available printing resources, enhance workflow transparency, and provide a more organized printing service. The overall objective is to establish an intelligent and coordinated printing environment that connects document processing, printer management, queue handling, payment processing, printing, and secure document collection within a single platform.

2.1 Specific Objectives

The specific objectives of IntelliPrint Nexus are:

1. To develop an AI-powered smart printing platform for managing campus printing activities through a centralized digital workflow.
2. To minimize printing waiting time by implementing intelligent queue management and waiting-time estimation based on relevant print-job and printer information.
3. To improve document quality by applying AI-based Computer Vision and document analysis techniques before printing.
4. To enable online document upload, print configuration, payment processing, and digital management of print requests.
5. To provide real-time print-job tracking, printer availability information, status updates, and notifications to improve transparency for users.
6. To enhance campus printing efficiency and user experience through intelligent printer management, automated workflow processing, and QR-based document collection.
7. To provide centralized administrative monitoring of users, print jobs, printers, queues, payments, notifications, and system analytics for efficient management of printing operations.
8. To maintain organized digital records of printing transactions and completed print jobs through database management, digital receipts, Print IDs, and QR Code-based document identification.

CHAPTER - 3

SYSTEM DEVELOPMENT METHODOLOGY

The development of IntelliPrint Nexus followed a structured system development methodology consisting of problem identification and requirement analysis, system design and architecture, Artificial Intelligence integration, application development and module integration, testing and validation, and user-interface refinement. These stages were carried out progressively to transform the identified campus printing problem into an integrated AI-powered printing management platform.The methodology provided a systematic approach for converting the identified requirements into functional and integrated system components. Based on these requirements, the system architecture and database structure were designed to support document processing, AI-based analysis, printer management, queue handling, payment processing, and print-job tracking. The development process focused on implementing and integrating these modules while ensuring proper communication between the application, AI components, and database.

3.1 Problem Identification and Requirement Analysis

The project began with the identification of limitations in conventional campus printing services. The existing process generally involved manual document submission, communication of printing requirements, payment verification, printer selection, queue handling, and physical collection of printed documents. During examinations, assignment deadlines, project submissions, and short academic breaks, increased printing demand can result in long queues, increased operator workload, inefficient printer utilization, and limited visibility of print-job status.

Based on these observations, the functional requirements of IntelliPrint Nexus were identified. Student requirements included secure authentication, document upload, printing preference selection, AI-assisted document analysis, printer availability information, payment processing, queue tracking, notifications, and QR-based document collection. Administrative requirements included management of students, printers, print jobs, payments, queues, notifications, reports, and system analytics. These requirements formed the basis for designing the system architecture and functional modules.

3.2 System Design and Architecture

Following requirement analysis, a modular architecture was designed to organize the major functions of IntelliPrint Nexus. The architecture separates authentication, student services, administrative services, AI processing, database management, queue management, printer management, payment processing, notifications, and supporting utilities. This separation provides clear responsibilities for individual components and supports systematic development and maintenance.

The architecture connects the modules through a common workflow. A student can submit a document, configure printing requirements, obtain AI-assisted analysis, complete payment, enter the printing queue, and monitor the job until completion. The administrative interface provides centralized access to printer, queue, user, payment, and print-job information.

3.3 Artificial Intelligence and Intelligent Automation

Artificial Intelligence was integrated to provide intelligent capabilities within the printing workflow. The Document Quality Analysis component uses Computer Vision techniques to examine uploaded documents before printing. OpenCV supports image-processing operations, while a CNN-based MobileNetV2 approach with transfer learning is incorporated for document quality assessment. The analysis considers characteristics such as sharpness, brightness, contrast, blur, orientation, and readability.

Machine Learning techniques are also incorporated into queue and printer management. Parameters such as document page count, queue length, printer availability, printer speed, workload, and job priority are considered for prediction-related tasks. Random Forest and XGBoost-based approaches support prediction and estimation activities, including waiting-time estimation and printer recommendation.

3.4 Application Development and Module Integration

The application was developed using Python and Streamlit, with SQLite and SQLAlchemy used for structured data management. The Student interface provides document upload, printing preferences, payment, queue tracking, notifications, print history, and QR-based document collection. The Administrative interface provides facilities for monitoring students, printers, print jobs, payments, queues, notifications, reports, and analytics.

The AI processing, queue management, printer management, and payment components were integrated into the application workflow. Supporting utilities were incorporated for PDF processing, QR Code generation, receipt generation, authentication, and printer-related operations. The modules were progressively integrated to establish a continuous workflow from document submission to final collection.

3.5 Testing, Validation, and System Optimization

After implementation, the major modules were tested individually and as an integrated system. Functional testing covered authentication, document upload, AI analysis, print configuration, payment processing, queue management, printer availability, print-job tracking, QR Code generation, notifications, and administrative functions. The complete workflow was also reviewed to verify the transfer of information between documents, print jobs, queues, printers, payments, and user interfaces.

Issues identified during development were addressed through debugging, code refinement, workflow adjustments, and interface improvements. Particular attention was given to the interaction between queue, printer, payment, and tracking components because these modules form the central operational workflow.

3.6 User Interface Refinement

The final stage focused on improving the usability, navigation, and consistency of the application. Interfaces were refined to provide clear access to document processing, AI analysis, printer information, payment, queue tracking, notifications, and administrative monitoring. The student workflow was organized from document submission and analysis through payment, queue processing, tracking, and collection, while the administrative interface provided centralized monitoring of system activities.

The completed modules were integrated into the final IntelliPrint Nexus platform, establishing an end-to-end digital printing workflow that combines software engineering, Artificial Intelligence, Computer Vision, Machine Learning, database management, queue processing, and printer management.

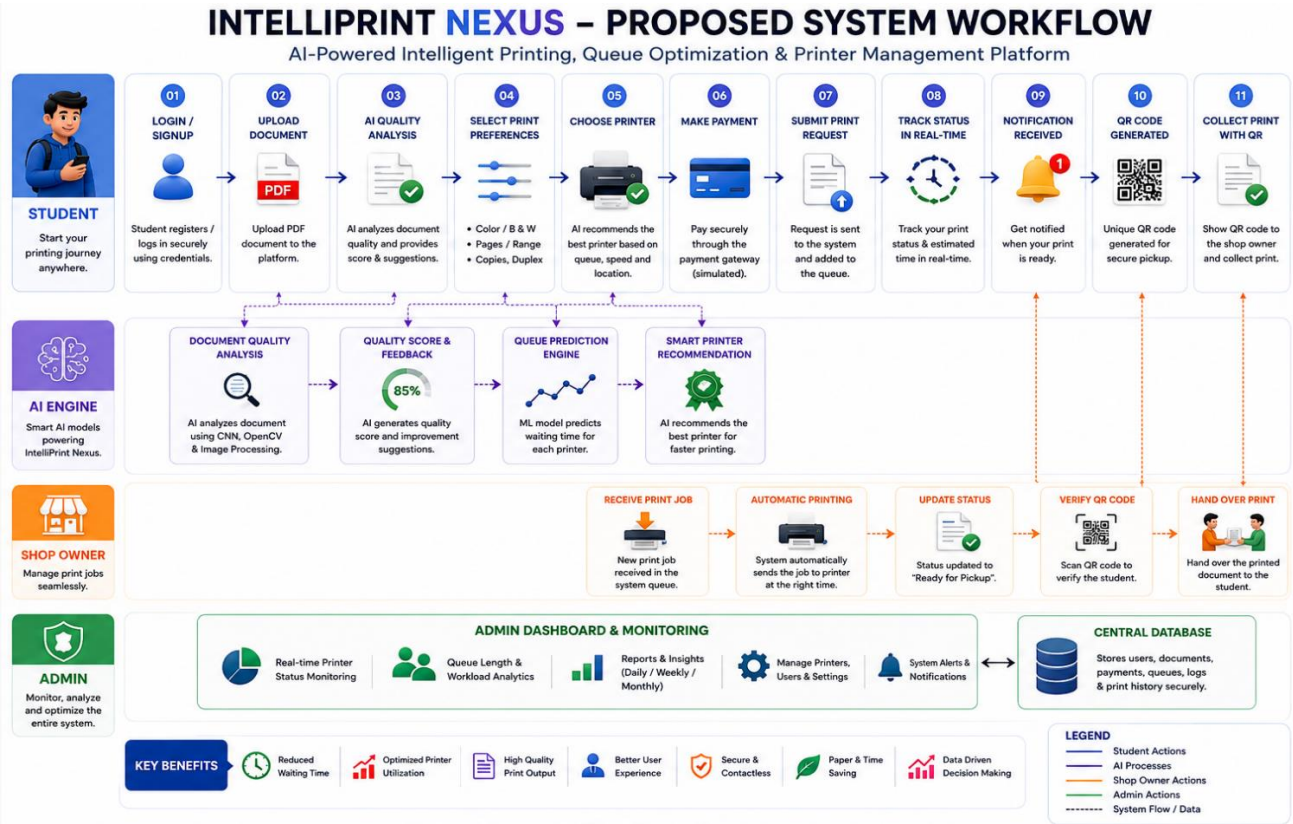


Figure 3.1: Overall Workflow of IntelliPrint Nexus

Above **Figure 3.1** presents the overall workflow of IntelliPrint Nexus, an AI-powered intelligent printing, queue optimization, and printer management platform. The workflow integrates student interaction, AI-based document analysis, intelligent printer recommendation, payment processing, print-job queuing, real-time status tracking, and secure document collection within a unified system. Initially, the student uploads a document and specifies the required printing preferences. The AI engine performs document quality analysis, generates a quality score, predicts the expected waiting time, and recommends a suitable printer based on queue conditions and printer availability. Following payment

and print-request submission, the job is added to the appropriate queue and processed automatically. The system continuously tracks the print status and generates a QR code for secure verification during document collection. In parallel, the shop owner can monitor and process print jobs, while the administrator can manage printers, users, reports, notifications, and system settings through the centralized database. Thus, the proposed workflow provides an automated and data-driven approach to printing management while improving queue efficiency, printer utilization, and the overall user experience.

CHAPTER - 4

SYSTEM DESIGN AND IMPLEMENTATION

4.1 Existing System Workflow

The existing campus printing process follows a manual workflow in which students depend on printing-shop operators for document submission, print configuration, payment, and printing. During peak periods such as examinations and assignment deadlines, large numbers of students create long queues and increased waiting times. As shown in **Figure. 4.1**, documents are generally shared through WhatsApp, e-mail, or USB devices, after which the operator manually downloads the files, confirms printing requirements such as copies, color mode, page range, and paper size, and processes the payment before printing.Since each request is handled individually, the operator must repeatedly communicate with students regarding printing preferences and payment details. This increases the workload of the printing-shop operator and may cause delays when several requests are received simultaneously. The absence of a centralized digital workflow also makes it difficult to maintain consistent records of printing requests and completed jobs.

The existing system also provides limited visibility into printer availability, queue position, estimated waiting time, and print-job progress. Students therefore have to remain at the printing shop and wait for their turn without real-time status information. The lack of automated queue management and intelligent printer allocation further reduces operational efficiency and results in a less convenient printing experience. These limitations demonstrate the need for an automated and intelligent campus printing system.

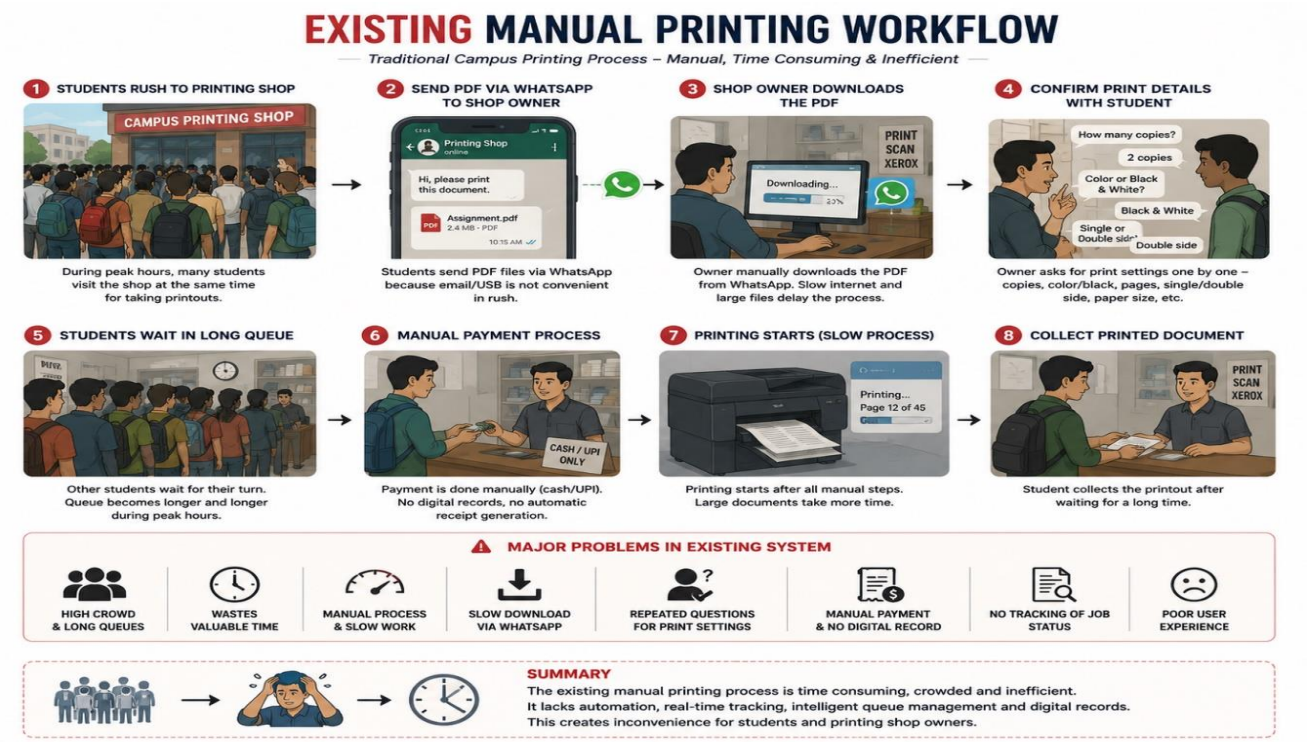


Figure 4.1: Existing Manual Campus Printing Workflow

4.2 Proposed IntelliPrint Nexus System Architecture

The proposed IntelliPrint Nexus architecture is designed as a modular and AI-powered printing platform that integrates students, print-shop administrators, Artificial Intelligence, a centralized database, and networked printers. As illustrated in **Figure 4.2**, the process begins when a student logs in, uploads a document, configures printing preferences, and submits a print request through the web application. The request is processed by the backend server and forwarded to the AI Engine for document quality assessment, estimated waiting-time prediction, queue optimization, and intelligent printer selection.

The web application provides separate interfaces for students and administrators. The student dashboard supports document upload, print settings, AI results, payment, queue tracking, and notifications, while the administrator dashboard provides access to incoming jobs, queue monitoring, printer management, payment verification, reports, and user management. The centralized database stores user information, documents, print jobs, payments, printer details, queues, notifications, and system records. The AI Engine uses CNN-based MobileNetV2 and OpenCV-based processing for document analysis and supports intelligent queue and printer management.

After successful payment, the print request is assigned to a suitable printer through the printer network, enabling automated processing and status updates. The system provides real-time information regarding the progress and estimated completion of print jobs. Once printing is completed, a unique QR Code and Print ID are generated for secure document collection. The proposed architecture reduces manual intervention, improves printer utilization, minimizes waiting time, and provides an integrated and efficient printing workflow for students and print-shop administrators.

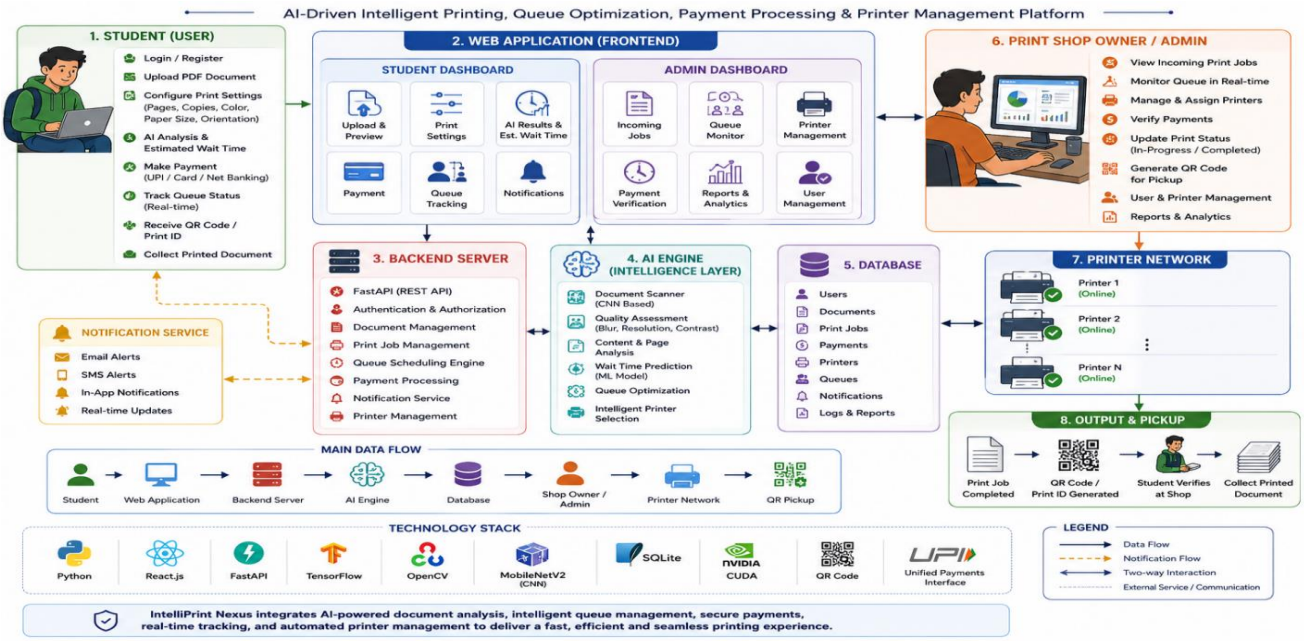


Figure 4.2: Proposed System Architecture of IntelliPrint Nexus

4.3 Technology Stack

The **IntelliPrint Nexus** system is developed using a modern full-stack technology ecosystem that combines web development, artificial intelligence, computer vision, database management, and GPU acceleration to provide a fast, secure, and intelligent printing platform. As illustrated in **Figure 4.3,** each technology has been selected based on its performance, scalability, reliability, and compatibility with AI-driven applications. The frontend provides an interactive user interface for students and administrators, while the backend manages authentication, document processing, queue scheduling, payment handling, and communication with the AI engine. Artificial Intelligence technologies are integrated to perform document quality assessment, printer recommendation, and intelligent queue optimization, whereas the database securely stores user information, print jobs, payment records, and printer details. Together, these technologies ensure seamless system integration and efficient execution of the complete printing workflow.

Table 4.1: Technologies Used in IntelliPrint Nexus

Technology	Purpose
Python	Primary programming language
Streamlit	Interactive application interface
SQLite	Database management
SQLAlchemy	Database interaction
TensorFlow/Keras	Deep-learning implementation
MobileNetV2	CNN-based document quality analysis
OpenCV	Image processing and computer vision
Scikit-learn	Machine-learning algorithms
XGBoost	Prediction and regression
NumPy	Numerical computation
Pandas	Data processing
QR Code Generator	Secure pickup identification
NVIDIA CUDA	GPU acceleration

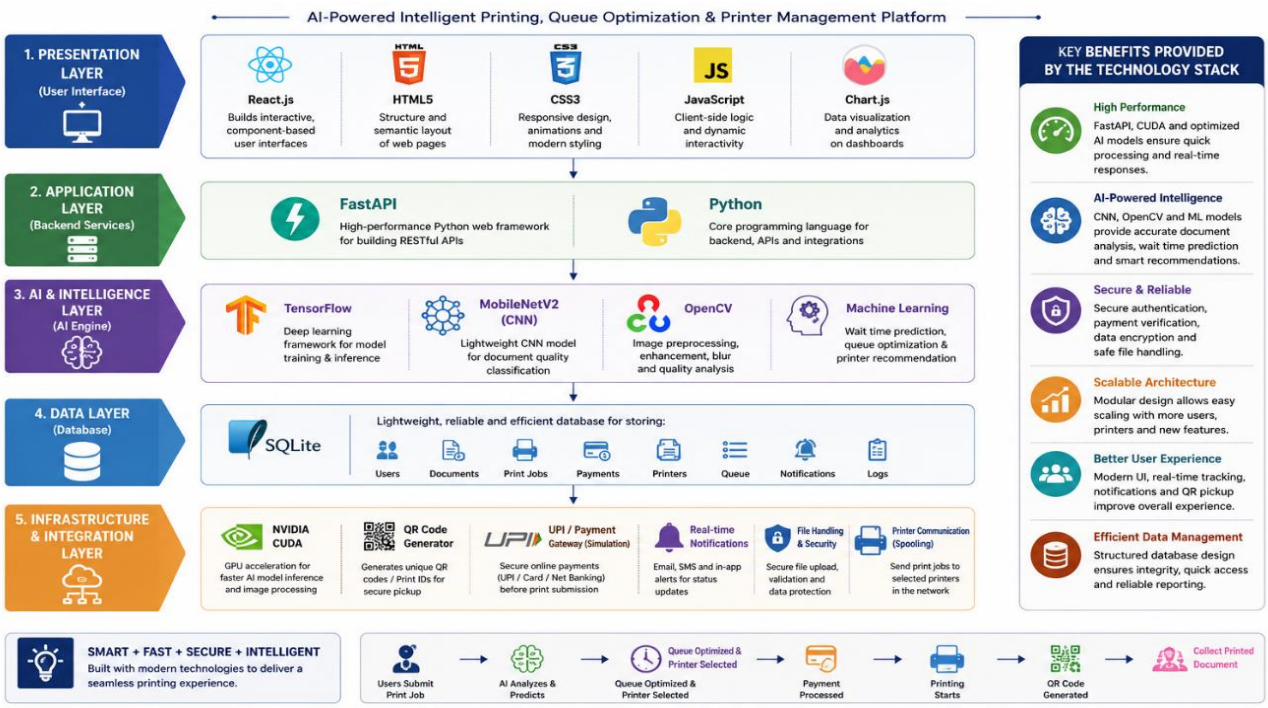


Figure 4.3: Technology Stack of IntelliPrint Nexus

4.3.1 Functional Capabilities Enabled by the Technology Stack

The integrated technology stack enables the following functional capabilities in the IntelliPrint Nexus system:

- **Responsive Web Application:** React.js and FastAPI support the development of interactive interfaces and communication between the frontend and backend services.
- **AI-Based Document Quality Analysis:** TensorFlow, MobileNetV2, and OpenCV support document analysis, image preprocessing, and quality assessment.
- **Intelligent Queue Prediction:** Machine Learning techniques are used to estimate waiting time and support queue optimization based on printing-related parameters.
- **Intelligent Printer Recommendation:** Printer information and estimated waiting time are considered to recommend a suitable printer for efficient print-job processing.
- **Payment Workflow Processing:** The system incorporates a simulated UPI-based payment workflow for processing printing payments before job submission.
- **Real-Time Dashboard Analytics:** Chart.js supports the visualization of printing statistics, system activity, printer information, and analytical data through dashboards.
- **QR Code-Based Document Pickup:** QR Code generation provides a unique identification mechanism for verifying and collecting completed print jobs.
- **GPU-Accelerated AI Processing:** NVIDIA CUDA support enables GPU-based acceleration for applicable AI model inference and computationally intensive image-processing operations.

4.4 AI Integration and Database Design

Artificial Intelligence and database management constitute the intelligent processing and data management components of the IntelliPrint Nexus platform. The AI Engine analyzes uploaded documents, estimates waiting time, and supports printer recommendation, while the database stores and manages user information, documents, print jobs, payment records, queue information, printer details, and notifications. The integration of these components enables automated decision-making and organized management of the printing workflow.

4.4.1 AI Integration

Artificial Intelligence forms a key component of IntelliPrint Nexus by providing automated document analysis and intelligent print management. After a document is uploaded, the AI Engine processes it using Convolutional Neural Network (CNN) techniques based on MobileNetV2 and OpenCV-based image processing. The document quality analysis evaluates characteristics such as blur, resolution, skewness, contrast, and readability. This analysis helps identify potential quality issues before the document is submitted for printing.

The AI Engine also considers operational parameters such as document page count, printer workload, queue length, printer speed, and printer availability for intelligent print management. Machine Learning techniques are used to estimate waiting time and support printer recommendation and queue optimization. These functions assist in distributing print jobs efficiently and reducing unnecessary delays during periods of high printing demand and reduce delays.

4.4.2 Major AI Functions

The major AI-related functions implemented in IntelliPrint Nexus are as follows:

- **Document Quality Assessment:** CNN-based MobileNetV2 and OpenCV techniques are used to analyze document quality.
- **Image Preprocessing:** OpenCV supports image processing and enhancement operations required for document analysis.
- **Waiting-Time Prediction:** Machine Learning techniques are used to estimate the expected waiting time based on printing-related parameters.
- **Queue Optimization:** The system uses queue-related information to support efficient scheduling and reduce unnecessary waiting.
- **Smart Printer Recommendation:** Printer availability, workload, speed, and queue information are considered for recommending a suitable printer.
- **Resource Optimization:** Intelligent processing supports improved utilization of available printers and efficient distribution of print jobs.

4.4.3 Database Design

The database layer provides centralized storage and management of the information generated throughout the IntelliPrint Nexus workflow. SQLite is used as the database system, with structured

data maintained for users, documents, print jobs, payments, printers, queues, notifications, and system records. The database enables different application modules to access and update relevant information during authentication, document submission, payment processing, queue management, printer allocation, and print-job tracking.

The database also supports the synchronization of operational information between the student interface, administrative dashboard, AI components, and printer-management modules. Print-job status, queue information, printer availability, and payment details can therefore be maintained as part of a common data-management structure. This centralized approach improves data consistency, simplifies information retrieval, and supports reliable operation of the complete printing workflow.

Table 4.2: Major Database Tables

Table Name	Purpose
Users	Stores student, faculty, administrator, and operator information.
Documents	Maintains uploaded document metadata and file details.
Print_Jobs	Records print requests, job status, and printer assignments.
Printers	Stores printer configuration, availability, toner level, and health status.
Queue	Maintains queue order, estimated waiting time, and scheduling information.
Payments	Stores transaction records and payment verification details.
Notifications	Records alerts, queue updates, payment confirmations, and print status notifications.
System_Logs	Maintains application events, user activities, and audit logs for monitoring and debugging.

4.4.4 Integration of AI and Database

The AI Engine and database operate together to support intelligent printing decisions. As illustrated in **Figure 4.4(a)**, information such as document properties, page count, queue length, printer status, and printer workload is obtained from the application and database and used by the AI components for analysis and prediction. The resulting quality assessment, estimated waiting time, and printer recommendation are then incorporated into the printing workflow. This integration allows the AI modules to utilize the operational data maintained by the database during the decision-making process. The availability of structured printing and printer information helps the system perform analysis based on the current conditions of the printing environment. It also provides a common information flow between the database, AI components, and application modules**.**

This integration enables IntelliPrint Nexus to connect AI-based decision-making with real-time printing operations. As illustrated in **Figure 4.4(b)**, it supports the complete process from document analysis and printer recommendation to queue assignment, printing, and status tracking, thereby providing an integrated approach to intelligent campus printing. The database maintains the information generated during these stages, allowing relevant updates to be accessed by the corresponding system components.

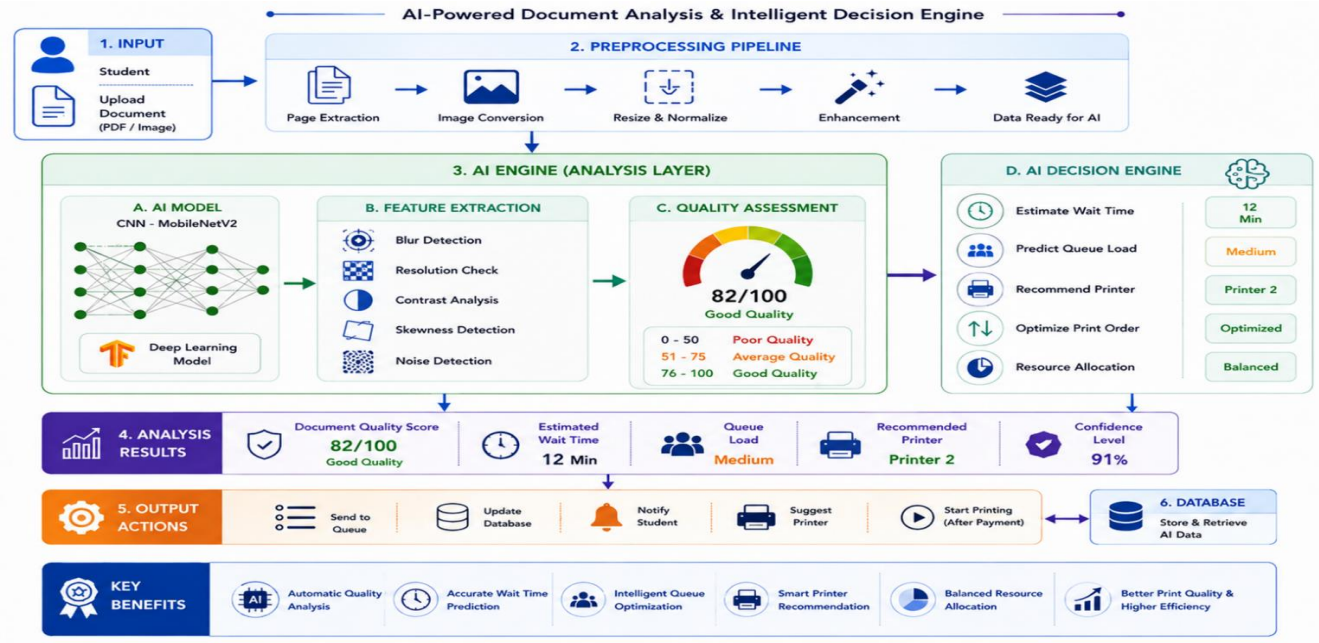


Figure 4.4(a): AI Processing Pipeline and Database Architecture of IntelliPrint Nexus

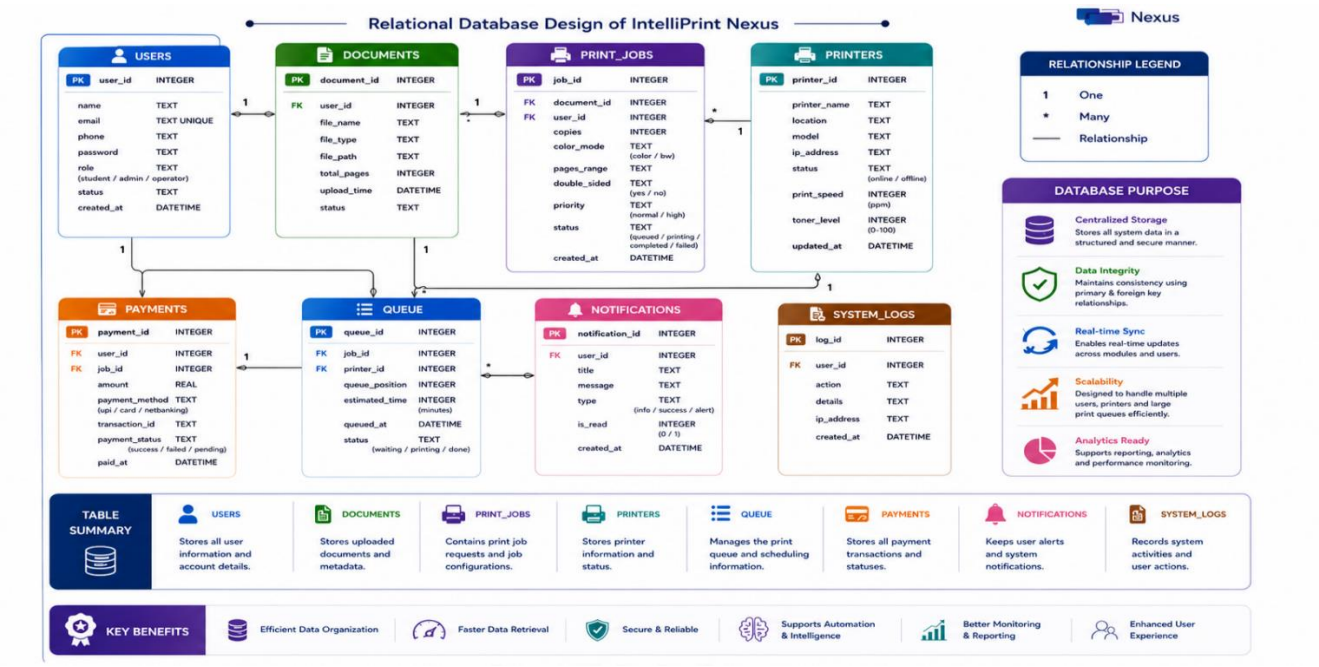


Figure 4.4(b): Database Schema

Observation

The integration of the Artificial Intelligence engine with the centralized database forms an important component of the IntelliPrint Nexus system, as it enables the platform to combine intelligent analysis with structured storage and retrieval of printing-related information. The AI engine utilizes information obtained from the application and database, including document properties, page count, queue length, printer status, and printer workload, to perform the required analysis and generate intelligent outputs. The document quality assessment module uses CNN-based MobileNetV2 and OpenCV-based image processing techniques to identify characteristics such as blur, resolution, skewness, contrast, and readability, thereby helping to identify potential document quality issues before printing.

In addition to document analysis, the integrated system supports intelligent print management through waiting-time prediction, queue optimization, and printer recommendation. The operational information maintained by the database provides the required input for estimating the expected waiting time and determining a suitable printer based on factors such as printer availability, workload, speed, and queue conditions. The resulting AI outputs are then incorporated into the printing workflow, allowing the system to support more efficient print-job allocation and reduce unnecessary delays. This integration establishes a connection between the intelligent processing layer and the operational printing environment rather than treating AI functions as independent modules.

The centralized database further provides a common data-management layer for maintaining information related to users, uploaded documents, print jobs, payments, printers, queues, notifications, and system logs. This enables the AI components and other application modules to access relevant and updated information during different stages of print-job processing. The database also supports synchronization between the student interface, administrative dashboard, AI engine, and printer-management modules, allowing changes in print-job status, queue information, printer availability, and payment details to be maintained consistently.

Overall, the integration of AI and database components enables IntelliPrint Nexus to establish a continuous information flow from document analysis and intelligent decision-making to printer recommendation, queue assignment, printing, and status tracking**.** The combined architecture reduces the separation between data management and intelligent processing, supports automated decision-making, improves the utilization of available printer resources, and provides a structured mechanism for managing printing operations. Therefore, the integration demonstrated in **Figure. 4.4(a)** and **Figure. 4.4(b)** provides the technical foundation for implementing an intelligent, centralized, and efficient campus printing management system.

CHAPTER - 5

RESULTS

The successful implementation of IntelliPrint Nexus demonstrates the practical application of Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Computer Vision (CV), and modern software engineering principles to address the operational challenges associated with conventional campus printing systems. The developed platform automates major stages of the document printing workflow by integrating intelligent decision-making, user management, payment processing, queue management, and real-time workflow monitoring into a unified digital platform. The system thereby transforms the conventional manual printing process into an automated and user-oriented solution.

The developed application successfully provides user authentication, PDF document upload, AI-assisted document quality analysis, intelligent queue prediction, printer recommendation, payment workflow simulation, QR Code-based document collection, receipt generation, and print-status tracking through an interactive interface. Separate student and administrative dashboards support the management of print jobs, printers, users, payments, notifications, and system information. The integration of these functions reduces manual intervention and improves the visibility and organization of printing operations.

The integration of AI-based analytical capabilities represents a major outcome of the project. Computer Vision and Deep Learning techniques are used for automated document quality assessment, while Machine Learning techniques support waiting-time prediction, printer recommendation, and queue-related decision-making. These functions utilize parameters such as document characteristics, printer availability, queue length, printer speed, and workload to support more efficient print-job processing. The resulting workflow improves printer utilization, reduces unnecessary delays, and provides users with better information regarding their printing requests.

The system also demonstrates the benefits of a modular software architecture in terms of reliability, maintainability, and integration. Individual components for authentication, document processing, AI analysis, database management, payment processing, printer management, queue tracking, notifications, and administrative monitoring can operate as coordinated modules within the overall platform. Structured database management and exception handling further support consistent system operation, while the user interface provides simplified access to the major functions of the application.

Overall, IntelliPrint Nexus demonstrates the feasibility of integrating Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, predictive analysis, and modern software engineering techniques to address a practical printing-management problem in educational institutions. The implemented system provides an integrated workflow covering document submission, quality analysis, printer recommendation, payment processing, queue management, printing, status tracking, and QR Code-based collection. The results indicate that the proposed platform can reduce dependence on manual processing, improve utilization of printing resources, and provide a more organized and convenient printing experience compared with the conventional printing workflow.

5.1 IntelliPrint Nexus Landing Page

The IntelliPrint Nexus Landing Page serves as the primary entry point to the proposed intelligent printing platform. It introduces the major capabilities of the system, including AI-based document analysis, smart queue prediction, printer recommendation, GPU-accelerated processing, secure payments, and real-time notifications. As shown in **Figure 5.1**, the interface presents the major technologies used in the platform, including Computer Vision, TensorFlow, OpenCV, Transfer Learning, and NVIDIA CUDA. The navigation options provide access to the main sections of the platform, while the intelligent workflow section presents the sequence from student login and document upload to AI analysis, printer recommendation, queue prediction, payment, and QR Code-based pickup.

The landing page also highlights key system metrics and benefits, including reduced waiting time, optimized printer utilization, resource savings, and improved user experience. The interface provides separate access points for users to log in or start using the platform and presents the overall purpose of IntelliPrint Nexus before entering the main application.



Figure 5.1: IntelliPrint Nexus Landing Page

The landing page provides an overview of the platform's major functionalities, intelligent workflow, technology stack, and expected operational benefits.

5.2 Student Dashboard

After successful authentication, the student is directed to the Student Dashboard, which serves as the central interface for managing printing activities. As shown in **Figure. 5.2**, the dashboard provides access to document upload, queue tracking, print history, analytics, notifications, payments, receipts, profile settings, and other printing services. It also displays key statistics such as total print jobs, documents printed, total pages, printing expenditure, completed jobs, average waiting time, and paper saved. These features enable students to monitor their printing activities and obtain relevant information from a single interface.

The dashboard further provides quick-action controls for frequently used operations and displays recent print jobs, printer status, monthly activity, and QR Code-based pickup information. The printer-status panel allows users to view the current state of available printers, while the recent-jobs section provides information about submitted printing requests. QR Code pickup codes support identification and collection of completed print jobs. By integrating these functions into a centralized interface, the Student Dashboard simplifies navigation, improves workflow visibility, and provides a convenient and organized printing experience.

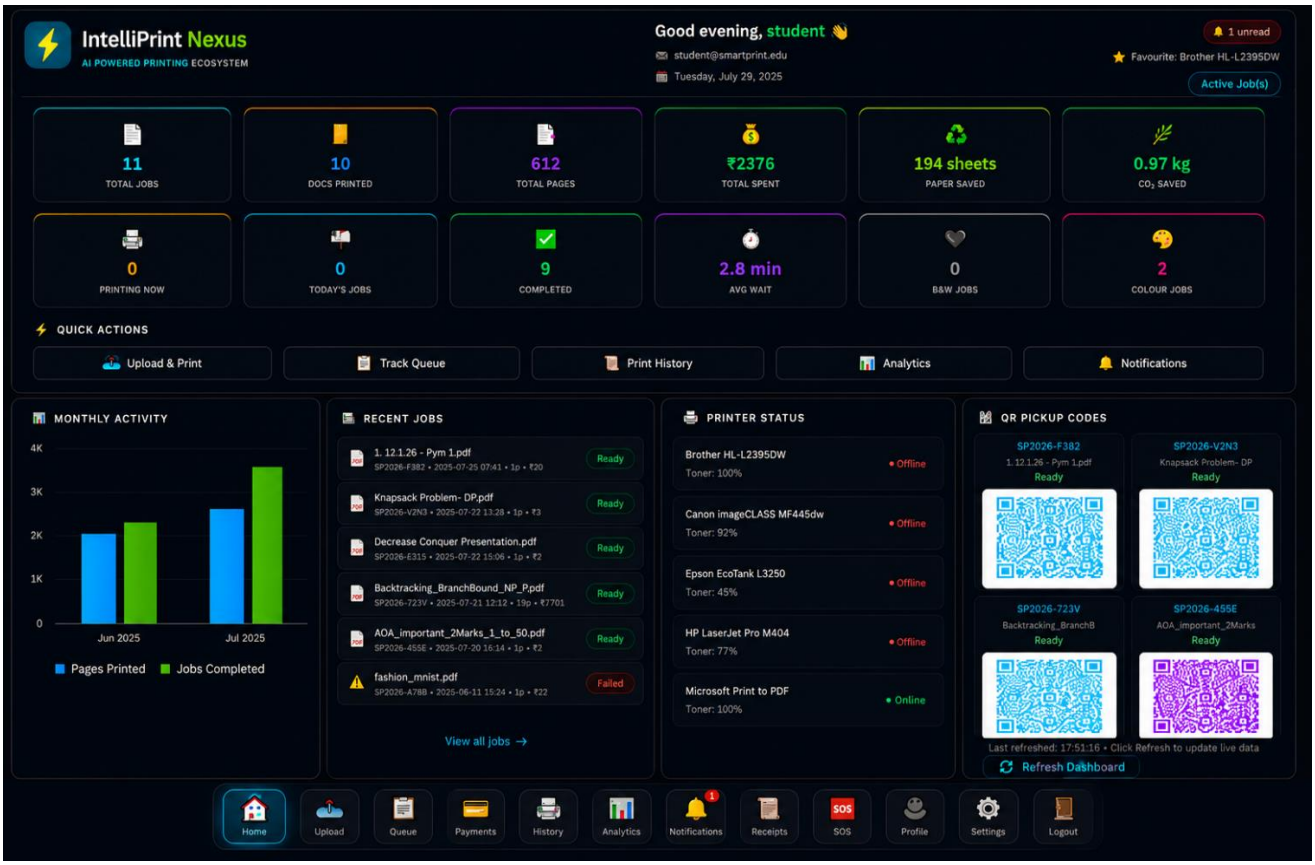


Figure 5.2. Student Dashboard.

The Student Dashboard provides centralized access to printing services and enables users to monitor print jobs, printer information, activity statistics, notifications, and QR Code-based document collection.

5.3 AI Document Quality Analysis

The AI Document Quality Analysis module is a major intelligent component of IntelliPrint Nexus that evaluates an uploaded document before printing. As shown in **Figure. 5.3**, the module provides a structured workflow beginning with document upload and print-setting configuration, followed by AI-based analysis and quality assessment. The system uses Computer Vision, OpenCV-based image processing, and a MobileNetV2-based Deep Learning model with transfer learning to examine document characteristics such as text clarity, layout structure, image sharpness, noise, and overall readability. The analysis generates a quality score and provides feedback regarding potential issues identified in the document.

Based on the analysis and available printer information, the system recommends a suitable printer and provides an estimated waiting time for the selected printer. The interface also presents relevant information such as printer availability, queue size, estimated wait time, printer speed, and toner status to support the selection process. The resulting quality assessment, printer recommendation, and waiting-time prediction assist users in making informed printing decisions before proceeding to payment and printing. This integrated AI workflow helps improve print reliability, reduce unnecessary reprinting and paper wastage, and support efficient utilization of available printers.



Figure. 5.3 AI Document Quality Analysis and Intelligent Printer Selection.

The module combines document quality assessment, intelligent printer recommendation, and waiting-time prediction to support informed and efficient printing decisions.

5.4 Payment and Queue Tracking

Following the completion of document analysis, IntelliPrint Nexus calculates the printing cost according to the document specifications and selected printing parameters. As shown in **Figure. 5.4**, the payment module allows the student to review order details, including the document, number of pages, copies, selected printer, estimated waiting time, and total amount before completing the payment. The system supports the configured payment workflow and records the transaction details associated with the print request. After successful payment, the print job proceeds to the queue for processing.

The Queue Tracking module provides continuous information about the submitted print job, including its current status, queue position, estimated completion time, printing progress, and selected printer. The tracking interface represents the different stages of the print workflow, from payment confirmation and queue assignment to preparation, printing, completion, and collection. This integration provides greater visibility into the printing process, reduces the need for repeated enquiries, and enables students to monitor their print requests until the document is ready for collection.

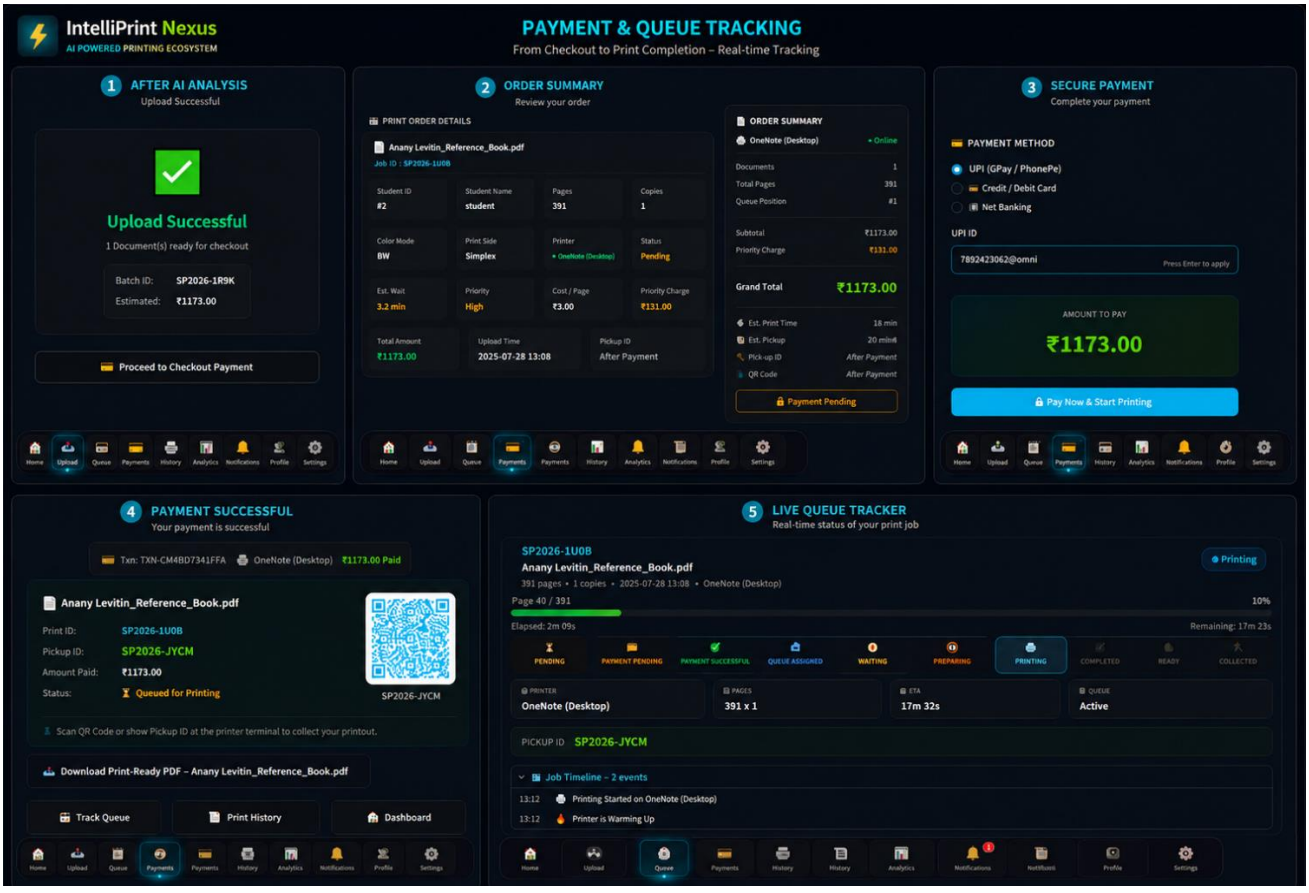


Figure. 5.4 Payment and Queue Tracking.

The integrated payment and queue-tracking modules support organized transaction processing and provide students with continuous visibility into the status and progress of their print jobs.

5.5 QR Pickup and Digital Receipt

After successful completion of the printing process, IntelliPrint Nexus generates a unique QR Code and Pickup ID for secure document collection. As shown in **Figure. 5.5**, the print-history interface provides users with access to job information, pickup details, downloadable receipts, and the original or print-ready document. The digital receipt contains relevant transaction information, including the print ID, pickup ID, document name, page count, print settings, selected printer, cost, total amount, and job status. The QR Code can be presented at the designated printer terminal to verify the print request before collection.

The QR-based pickup mechanism provides a convenient method for identifying completed print jobs and reduces dependence on manual verification. The job timeline also provides information about important stages such as document upload, payment confirmation, queue assignment, and print readiness. By combining QR-based identification, digital receipts, print history, and job-status information, the module improves transparency and organization during document collection while maintaining a digital record of completed printing activities.



Figure. 5.5 QR Pickup and Digital Receipt.

The QR-based document collection module provides unique print-job identification, digital receipt generation, and access to printing history for organized and convenient document pickup.

5.6 Admin Dashboard and Analytics

The Admin Dashboard provides a centralized interface for monitoring and managing the IntelliPrint Nexus printing ecosystem. As shown in **Figure. 5.6**, the dashboard presents key operational information including revenue, daily documents and pages, average waiting time, print jobs, completed jobs, registered students, and printer availability. It also provides dedicated sections for queue management, printer management, print jobs, analytics, student management, reports, activity logs, and system settings. These integrated functions enable administrators to monitor printing operations and access important system information from a single interface.

The dashboard further provides analytical information through job-status distribution, recent activity, performance trends, revenue statistics, printer-terminal status, and consumable and printer-health monitoring. The interface also displays AI hardware and system-resource information, including CPU, GPU, memory, and related utilization indicators where available. By combining operational monitoring with analytics and resource information, the Admin Dashboard supports informed management of users, print jobs, queues, printers, and overall system performance.



Figure. 5.6 Admin Dashboard and Analytics.

The Admin Dashboard provides centralized monitoring, operational analytics, printer and resource status, and management controls for the IntelliPrint Nexus platform.

CHAPTER - 6

CONCLUSION

6.1 Conclusion

The successful development of IntelliPrint Nexus demonstrates the practical integration of Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Computer Vision (CV), predictive analytics, and modern software engineering to address challenges associated with conventional campus printing services. The system transforms the traditional manual printing process into an organized digital workflow by integrating document submission, document quality analysis, intelligent queue management, printer recommendation, payment processing, print-job tracking, and QR Code-based document collection. These capabilities help improve workflow transparency, reduce unnecessary manual intervention, and support more efficient utilization of available printing resources.

A significant achievement of the system is the implementation of AI-based Document Quality Analysis using Computer Vision, image-processing techniques, and transfer learning with the MobileNetV2 architecture. The system analyzes uploaded documents before printing and identifies potential quality-related issues that may affect the final output. In addition, Machine Learning-based queue prediction and intelligent printer recommendation consider relevant parameters such as document page count, queue length, printer availability, printer speed, workload, and job priority to support waiting-time estimation and suitable printer selection. These intelligent capabilities contribute to better print-job management and improved decision-making within the printing workflow.

The developed platform integrates user authentication, document upload, print configuration, payment processing, QR Code-based document collection, queue tracking, digital receipt generation, notifications, administrative monitoring, and printer management within a unified architecture. The modular organization of the system supports maintainability and allows individual components to be modified or extended independently. The use of Python, Streamlit, SQLite, SQLAlchemy, TensorFlow, Keras, OpenCV, Scikit-learn, XGBoost, NumPy, Pandas, and supporting libraries demonstrates the practical application of modern software development and Artificial Intelligence technologies in developing an intelligent campus printing platform.

The project also provided practical exposure to the software development lifecycle, including requirement analysis, system design, application development, AI integration, database management, module integration, testing, debugging, and user-interface refinement. The developed system demonstrates the feasibility of applying intelligent automation and predictive technologies to campus printing operations and provides a structured foundation for improving printing efficiency, workflow transparency, resource utilization, and user experience.

6.2 Future Enhancements

Although IntelliPrint Nexus provides an integrated solution for intelligent campus printing management, several enhancements can be considered for future development. The system can be

extended through cloud deployment to support centralized access and scalability across multiple campus locations. Integration with IoT-enabled smart printers can provide real-time printer status, toner-level monitoring, paper-level monitoring, and automated fault detection.

The simulated payment workflow can be extended to support real-time digital payment integration using appropriate secure payment gateways. Predictive maintenance techniques can also be incorporated to identify potential printer failures before they occur and reduce service interruptions. A dedicated mobile application can further improve accessibility by allowing students and administrators to manage printing activities through mobile devices.

Future versions can additionally incorporate Large Language Model (LLM)-based assistants to provide natural-language support for printing-related queries, system navigation, and troubleshooting. Advanced AI analytics can be introduced to analyze historical printing patterns, predict demand, optimize printer allocation, and provide improved operational insights. These enhancements can further improve the scalability, intelligence, automation, and practical applicability of IntelliPrint Nexus in larger institutional environments.

6.3 Skills and Technical Learning

The development of IntelliPrint Nexus provided practical experience in several areas of software engineering and Artificial Intelligence. The project strengthened skills in Python programming, web application development, database management, Machine Learning, Deep Learning, Computer Vision, AI model integration, system architecture, and software testing. Practical exposure was also gained in integrating different software components, managing application workflows, debugging implementation issues, and developing user-oriented interfaces.

The internship further developed skills in problem identification, requirement analysis, technical documentation, system design, modular development, testing, and project-based problem solving. Working with technologies such as TensorFlow, OpenCV, Scikit-learn, XGBoost, SQLite, SQLAlchemy, and Streamlit provided practical understanding of how modern technologies can be combined to develop an end-to-end intelligent software solution.

Overall, the project provided valuable practical experience in transforming a real-world problem into a structured software solution and strengthened the technical and problem-solving skills required for future work in Artificial Intelligence, Machine Learning, software engineering, and intelligent application development.

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CHAPTER – 8

APPENDICES

APPENDIX 1

Source Snippets

7.1 Frontend / User Interface

```
with st.container():
    col_file, col_settings = st.columns([1.2, 1])

    with col_file:
        uploaded_files = st.file_uploader(
            "Choose PDF files (Max 10)",
            type=["pdf"],
            accept_multiple_files=True,
            help="Upload multiple PDFs at once.",
            key="pdf_uploader"
        )

    with col_settings:
        global_copies = st.number_input(
            "Copies",
            min_value=1,
            max_value=100,
            value=1
        )

        global_color = st.selectbox(
            "Color Mode",
            ["Black & White", "Color"],
            index=0
        )

        global_duplex = st.selectbox(
            "Print Sides",
            ["Single-sided", "Double-sided"],
            index=0
        )

        priority = st.toggle(
            " Priority Print (+50% cost)",
            value=False
        )
```

7.2 AI-Based Document Analysis & Quality Assessment

```
def _build_document_classifier():
    """Build CNN using Transfer Learning
    (MobileNetV2 base, custom classification head).
```

```
    """

    base_model = MobileNetV2(
        weights="imagenet",
        include_top=False,
        input_shape=(_IMAGE_SIZE, _IMAGE_SIZE, 3),
    )

    base_model.trainable = False

    x = GlobalAveragePooling2D()(
        base_model.output
    )

    x = Dense(
        128,
        activation="relu"
    )(x)

    x = Dropout(0.5)(x)

    predictions = Dense(
        len(CLASSES),
        activation="softmax"
    )(x)

    model = Model(
        inputs=base_model.input,
        outputs=predictions
    )
```

7.3 Intelligent Queue Prediction & Printer Recommendation

```
pred = predict_wait_time(
    queue_length=queue_len,
    pages=job_pages,
    copies=job_copies,
    printer_speed_ppm=speed,
    priority_flag=False,
)

wait_time = float(
    pred["estimated_wait_minutes"]
)
```

```
score -= wait_time * 4.0

scored_printers.append({
    "printer_id": p.get("id"),
    "name": p.get("name", "Printer"),
    "score": round(
        max(0.0, min(100.0, score)), 2
    ),
    "ppm": speed,
    "queue": queue_len,
    "predicted_wait": wait_time,
})

scored_printers.sort(
    key=lambda x: x["score"],
    reverse=True
)
```

7.4 Database & Print-Job Management

```
job = PrintJob(
    user_id=user_id,
    document_name=document_name,
    file_path=file_path,
    print_id=print_id,
    status="Pending",
    copies=copies,
    color_mode=color_mode,
    duplex=duplex,
    paper_size=paper_size,
    priority=priority,
    total_pages=total_pages,
    total_cost=total_cost,
    quality_score=quality_score,
    printer_id=printer_id,
    qr_code_path=qr_code_path,
    estimated_wait_time=estimated_wait_time,
    submitted_at=datetime.utcnow(),
)

db.add(job)
db.commit()
db.refresh(job)
return _job_to_dict(job)
```

APPENDIX 2

DAILY WORK

Day 1: Environment Setup and GPU Infrastructure

The first day of the internship focused on setting up the development environment required for GPU-accelerated Artificial Intelligence and High-Performance Computing tasks. A GPU-enabled workspace was configured at the NVIDIA AI Center of Excellence, and the required development dependencies and CUDA environment were prepared. SSH was used to access the remote computing environment, while Kubernetes commands were explored for managing and deploying containerized resources. Port mapping was configured to enable access to services running within the containers, and Jupyter Notebook was launched through a web browser for interactive development. This session provided practical exposure to GPU-enabled infrastructure and demonstrated how containerization, resource allocation, and GPU computing can support computationally intensive AI workloads.

Day 2: Machine Learning Foundations

The second day focused on the fundamentals of classical machine learning and statistical inference. Several supervised and unsupervised learning algorithms were implemented and evaluated using Python. The experiments included Linear Regression, Logistic Regression, Polynomial Regression, Decision Trees, Random Forests, Support Vector Machines, Naive Bayes, K-Means Clustering, and Principal Component Analysis. OpenML datasets were used to support reproducible experimentation. The session also involved continuous target prediction, binary classification, dimensionality reduction, and feature-space visualization. Different impurity measures, including Gini impurity and Entropy, were compared for decision-tree models. These activities provided a strong foundation in machine learning algorithms, model behavior, feature representation, and statistical reasoning.

Day 3: Neural Networks from Scratch

The third day introduced the fundamental concepts of deep learning through the implementation of neural network components from first principles. Forward propagation, backpropagation, and gradient descent optimization were studied and implemented. Different activation functions, including Sigmoid and ReLU, were explored to understand their contribution to nonlinear learning. Loss convergence was analyzed during the training process to understand how model parameters are optimized. Implementing the fundamental components manually provided a deeper understanding of the mathematical operations involved in neural network training and established the foundation for working with modern deep learning frameworks.

Day 4: Convolutional Neural Networks and Profiling

The fourth day focused on Computer Vision and Convolutional Neural Networks (CNNs). The architecture and working principles of CNNs were studied, with particular emphasis on convolution operations, spatial feature extraction, and pooling. A three-layer CNN architecture was implemented

and used to investigate how hierarchical features are extracted from image data. Performance profiling was also performed to understand the computational requirements of deep learning operations. The session demonstrated the importance of efficient computation and GPU utilization when training and executing computer vision models.

Day 5: Model Architecture Benchmarking

The fifth day involved benchmarking different computer vision architectures using the Intel Image Dataset. A custom CNN architecture was compared with established pre-trained models, including MobileNetV2, ResNet50, and EfficientNetB0. The models were evaluated based on accuracy, training efficiency, inference throughput, and model complexity. The comparison provided practical insight into the trade-offs between computational cost, model performance, and inference speed. The experiments also demonstrated the benefits of using pre-trained architectures for computer vision tasks and highlighted the importance of selecting an appropriate model based on the requirements of a particular application.

Day 6: Transformer Architecture and GPU Analysis

The sixth day focused on Transformer architectures and their role in modern Natural Language Processing and Large Language Models. The internal components of Transformer models were studied, including token embeddings and attention mechanisms. The self-attention mechanism was examined to understand how contextual relationships between tokens are captured. GPU memory utilization, VRAM consumption, system RAM usage, and model throughput were also monitored during experimentation. This session provided an understanding of the computational requirements of Transformer-based models and highlighted the importance of hardware-aware optimization when working with large-scale AI models.

Day 7: Parameter-Efficient Fine-Tuning Using LoRA

The seventh day focused on Parameter-Efficient Fine-Tuning (PEFT), with particular emphasis on Low-Rank Adaptation (LoRA). LoRA was implemented using the DistilGPT-2 model to study efficient adaptation of Transformer-based language models. The configuration used a LoRA rank of 8 and an alpha value of 16. The experiments examined the reduction in trainable parameters and memory requirements compared with conventional fine-tuning approaches. The session demonstrated how PEFT techniques can reduce computational requirements while allowing large language models to be adapted for specific tasks. This approach enables faster experimentation and makes model fine-tuning more accessible on limited computational resources.

Day 8: Generative AI and Diffusion Models

The eighth day introduced generative AI concepts through the implementation of a Denoising Diffusion Probabilistic Model (DDPM) using PyTorch. A custom U-Net architecture consisting of downsampling and upsampling components was developed for the diffusion process. A linear noise scheduler with 1,000 diffusion timesteps was implemented, followed by training and sampling

workflows. The experiments provided practical understanding of how noise is progressively introduced into data and subsequently removed through an iterative denoising process. This session strengthened the understanding of modern generative architectures and their computational requirements.

Day 9: GPU Optimization and Multi-Modal AI Inference

The ninth day focused on modern AI inference systems and computational optimization. Multi-modal AI workflows involving text generation and image understanding were explored. Prompt benchmarking and context-window optimization were performed to understand their impact on inference behavior and resource utilization. Several AI service integrations were explored using Google GenAI, OpenAI, and Groq SDKs. The session also introduced production-oriented AI workflows and multi-modal orchestration. These activities demonstrated the importance of efficient model selection, prompt design, context management, API integration, and GPU-aware optimization when developing practical generative AI applications.

Day 10: Internship Conclusion and Project Discussions

The final day of the internship focused on reviewing the technical concepts covered throughout the program and discussing potential capstone project ideas. Discussions were conducted regarding possible AI-based project concepts and their practical implementations. The major technical topics covered during the internship were reviewed to consolidate the knowledge gained throughout the training period. A summary of the internship learning outcomes was prepared, highlighting the key concepts, tools, technologies, and practical skills acquired during the program. The key technical learnings and observations were presented to the mentors for review. The internship concluded with a final review and feedback session, where valuable suggestions and guidance were received regarding future project development and the practical application of the concepts learned during the internship.

Overall Work Progress

The internship followed a progressive learning path, beginning with GPU infrastructure and classical machine learning and advancing toward deep learning, computer vision, Transformer architectures, parameter-efficient fine-tuning, generative AI, and multi-modal inference. The practical sessions provided exposure to both fundamental concepts and modern AI engineering practices. The overall progression from machine learning algorithms to advanced generative AI systems helped establish an understanding of the complete AI development workflow, including environment configuration, model implementation, training, evaluation, optimization, inference, and project-level integration.